Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Full Circle Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
6800 East AVE	Ballico	Merced	95303

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/2001

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X042-X040-X028-XXXX X042-X040-X029-XXXX X042-X040-X034-XXXX

B. OPERATORS

Bylsma, Brian

Operator name: Bylsma, Brian

Telephone no.: (209) 605-0235

Mailing Address Number and Street	City	State	Zip Code
6800 East AVE	Ballico	CA	95303

This operator is responsible for paying permit fees.

C. OWNERS

Burroughs, Ward

Legal owner name: Burroughs, Ward

Telephone no.: (209) 605-0235

Mailing Address Number and Street	City	State	Zip Code
22329 E Monte Vista AVE	Denair	CA	95316

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	80	40	0	0
Number under roof	245	40	0	0	0	0
Maximum number	550	40	100	50	0	0
Average number	245	40	80	40	0	0
Avg live weight (lbs)	1,000	1,100	725	600		

Predominant milk cow breed: Jersey-Holstein Cross

Average milk production: 40 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 7,074.56 *tons per reporting period*

Total nitrogen from manure: 89,577.84 *lbs per reporting period*

After ammonia losses (30% loss applied): 62,704.49 *lbs per reporting period*

Total phosphorus from manure: 14,727.60 *lbs per reporting period*

Total potassium from manure: 42,090.81 *lbs per reporting period*

Total salt from manure: 124,556.25 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 6,273,000 *gallons*

Total nitrogen generated: 28,520.82 *lbs*

Total phosphorus generated: 3,812.99 *lbs*

Total potassium generated: 38,787.45 *lbs*

Total salt generated: 167,437.58 *lbs*

	6,273,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	6,273,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
IW Field 4	Ground water
IW PVT 1	Ground water
IW PVT 2 & 3	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 4	43	43	1	process wastewater	X042-X040-X034-XXXX
Field 5	9	9	1	process wastewater	X042-X040-X034-XXXX
Native Pasture #1	268	268	0	none	X042-X040-X033-XXXX X042-X040-X034-XXXX
Native Pasture #2	128	128	0	none	X042-X040-X028-XXXX X042-X040-X029-XXXX
PVT 1	125	125	1	both	X042-X040-X034-XXXX
PVT 2	95	95	1	manure	X042-X040-X028-XXXX
PVT 3	95	95	1	manure	X042-X040-X029-XXXX
Totals for areas that were used for application	367	367	5		
Totals for areas that were not used for application	396	396	0		
Land application area totals	763	763	5		

B. CROPS AND HARVESTS

Field 4

Field name: Field 4

01/01/2023: Pasture

Crop: Pasture Acres planted: 43 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	645.00 *ton*	As-is		70.0	9,180.00	1,150.00	10,900.00		14.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	276.00	39.00	283.50	0.00
Total actual harvest content	15.00	275.40	34.50	327.00	1,305.00

Field 5

Field name: Field 5

Field 5

01/01/2023: Pasture

Crop: Pasture Acres planted: 9 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	135.00 *ton*	As-is		70.0	9,730.00	1,100.00	11,300.00		14.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	276.00	39.00	283.50	0.00
Total actual harvest content	15.00	291.90	33.00	339.00	1,287.00

PVT 1

Field name: PVT 1

01/01/2023: Pasture

Crop: Pasture Acres planted: 125 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	1,875.00 *ton*	As-is		70.0	11,800.00	650.00	7,280.00		14.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	276.00	39.00	283.50	0.00
Total actual harvest content	15.00	354.00	19.50	218.40	1,314.00

PVT 2

Field name: PVT 2

PVT 2

01/01/2023: Pasture

Crop: Pasture Acres planted: 95 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	1,425.00 *ton*	As-is		70.0	12,800.00	1,100.00	14,100.00		14.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	276.00	39.00	283.50	0.00
Total actual harvest content	15.00	384.00	33.00	423.00	1,341.00

PVT 3

Field name: PVT 3

01/01/2023: Pasture

Crop: Pasture Acres planted: 95 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	1,425.00 *ton*	As-is		70.0	13,300.00	1,160.00	14,900.00		15.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	276.00	39.00	283.50	0.00
Total actual harvest content	15.00	399.00	34.80	447.00	1,386.00

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 4 - 01/01/2023: Pasture

Field name: Field 4

Crop: Pasture

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Field 4	Ground water	3.58	0.00	0.00	55.33	5,184,000.00 *gal*
Application event totals		3.58	0.00	0.00	55.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Field 4	Ground water	3.28	0.00	0.00	50.72	4,752,000.00 *gal*
Application event totals		3.28	0.00	0.00	50.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	76.79	14.97	150.36	726.88	918,000.00 *gal*
IW Field 4	Ground water	3.13	0.00	0.00	48.42	4,536,000.00 *gal*
Application event totals		79.92	14.97	150.36	775.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	51.19	9.98	100.24	484.58	612,000.00 *gal*
IW Field 4	Ground water	3.13	0.00	0.00	48.42	4,536,000.00 *gal*
Application event totals		54.32	9.98	100.24	533.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Field 4	Ground water	2.91	0.00	0.00	44.96	4,212,000.00 *gal*
Application event totals		2.91	0.00	0.00	44.96	

Field 4 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Field 4	Ground water	2.84	0.00	0.00	43.81	4,104,000.00 *gal*
Application event totals		2.84	0.00	0.00	43.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Field 4	Ground water	2.69	0.00	0.00	41.50	3,888,000.00 *gal*
Application event totals		2.69	0.00	0.00	41.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	93.83	5.80	61.64	154.40	612,000.00 *gal*
IW Field 4	Ground water	2.69	0.00	0.00	41.50	3,888,000.00 *gal*
Application event totals		96.52	5.80	61.64	195.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Field 4	Ground water	2.69	0.00	0.00	41.50	3,888,000.00 *gal*
Application event totals		2.69	0.00	0.00	41.50	

Field 5 - 01/01/2023: Pasture

Field name: Field 5

Crop: Pasture

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Field 4	Ground water	4.28	0.00	0.00	66.09	1,296,000.00 *gal*
Application event totals		4.28	0.00	0.00	66.09	

Field 5 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Field 4	Ground water	3.92	0.00	0.00	60.58	1,188,000.00 *gal*
Application event totals		3.92	0.00	0.00	60.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	61.14	11.92	119.73	578.81	153,000.00 *gal*
IW Field 4	Ground water	3.56	0.00	0.00	55.08	1,080,000.00 *gal*
Application event totals		64.71	11.92	119.73	633.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	61.14	11.92	119.73	578.81	153,000.00 *gal*
IW Field 4	Ground water	3.21	0.00	0.00	49.57	972,000.00 *gal*
Application event totals		64.35	11.92	119.73	628.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Field 4	Ground water	2.85	0.00	0.00	44.06	864,000.00 *gal*
Application event totals		2.85	0.00	0.00	44.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Field 4	Ground water	2.85	0.00	0.00	44.06	864,000.00 *gal*
Application event totals		2.85	0.00	0.00	44.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Field 4	Ground water	2.85	0.00	0.00	44.06	864,000.00 *gal*
Application event totals		2.85	0.00	0.00	44.06	

Field 5 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	112.07	6.92	73.63	184.42	153,000.00 *gal*
IW Field 4	Ground water	2.85	0.00	0.00	44.06	864,000.00 *gal*
Application event totals		114.93	6.92	73.63	228.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Field 4	Ground water	2.85	0.00	0.00	44.06	864,000.00 *gal*
Application event totals		2.85	0.00	0.00	44.06	

PVT 1 - 01/01/2023: Pasture

Field name: PVT 1

Crop: Pasture

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW PVT 1	Ground water	0.00	0.00	0.00	42.45	12,717,900.00 *gal*
Application event totals		0.00	0.00	0.00	42.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.22	6.86	68.97	333.39	1,224,000.00 *gal*
IW PVT 1	Ground water	0.00	0.00	0.00	45.72	13,696,200.00 *gal*
Application event totals		35.22	6.86	68.97	379.11	

PVT 1 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.22	6.86	68.97	333.39	1,224,000.00 *gal*
IW PVT 1	Ground water	0.00	0.00	0.00	46.81	14,022,300.00 *gal*
Application event totals		35.22	6.86	68.97	380.20	

05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW PVT 1	Ground water	0.00	0.00	0.00	45.72	13,696,200.00 *gal*
Application event totals		0.00	0.00	0.00	45.72	

06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW PVT 1	Ground water	0.00	0.00	0.00	45.72	13,696,200.00 *gal*
Application event totals		0.00	0.00	0.00	45.72	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW PVT 1	Ground water	0.00	0.00	0.00	45.72	13,696,200.00 *gal*
Application event totals		0.00	0.00	0.00	45.72	

08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW PVT 1	Ground water	0.00	0.00	0.00	46.81	14,022,300.00 *gal*
Application event totals		0.00	0.00	0.00	46.81	

09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	64.55	3.99	42.41	106.23	1,224,000.00 *gal*
IW PVT 1	Ground water	0.00	0.00	0.00	46.81	14,022,300.00 *gal*
Application event totals		64.55	3.99	42.41	153.03	

PVT 1 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	142.72	41.75	373.93	1,940.99	500.00 *ton*
Application event totals		142.72	41.75	373.93	1,940.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW PVT 1	Ground water	0.00	0.00	0.00	45.72	13,696,200.00 *gal*
Application event totals		0.00	0.00	0.00	45.72	

PVT 2 - 01/01/2023: Pasture

Field name: PVT 2

Crop: Pasture

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW PVT 2 & 3	Ground water	2.85	0.00	0.00	30.97	8,814,000.00 *gal*
Application event totals		2.85	0.00	0.00	30.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW PVT 2 & 3	Ground water	2.71	0.00	0.00	29.42	8,373,300.00 *gal*
Application event totals		2.71	0.00	0.00	29.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW PVT 2 & 3	Ground water	2.71	0.00	0.00	29.42	8,373,300.00 *gal*
Application event totals		2.71	0.00	0.00	29.42	

PVT 2 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	254.94	53.20	21.06	3,014.91	750.00 *ton*
Application event totals		254.94	53.20	21.06	3,014.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW PVT 2 & 3	Ground water	3.13	0.00	0.00	34.07	9,695,400.00 *gal*
Application event totals		3.13	0.00	0.00	34.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW PVT 2 & 3	Ground water	2.99	0.00	0.00	32.52	9,254,700.00 *gal*
Application event totals		2.99	0.00	0.00	32.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW PVT 2 & 3	Ground water	2.85	0.00	0.00	30.97	8,814,000.00 *gal*
Application event totals		2.85	0.00	0.00	30.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW PVT 2 & 3	Ground water	2.71	0.00	0.00	29.42	8,373,300.00 *gal*
Application event totals		2.71	0.00	0.00	29.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW PVT 2 & 3	Ground water	2.71	0.00	0.00	29.42	8,373,300.00 *gal*
Application event totals		2.71	0.00	0.00	29.42	

PVT 2 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW PVT 2 & 3	Ground water	2.56	0.00	0.00	27.87	7,932,600.00 *gal*
Application event totals		2.56	0.00	0.00	27.87	

PVT 3 - 01/01/2023: Pasture

Field name: PVT 3

Crop: Pasture

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW PVT 2 & 3	Ground water	3.28	0.00	0.00	35.62	10,136,100.00 *gal*
Application event totals		3.28	0.00	0.00	35.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW PVT 2 & 3	Ground water	3.13	0.00	0.00	34.07	9,695,400.00 *gal*
Application event totals		3.13	0.00	0.00	34.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW PVT 2 & 3	Ground water	3.13	0.00	0.00	34.07	9,695,400.00 *gal*
Application event totals		3.13	0.00	0.00	34.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW PVT 2 & 3	Ground water	2.99	0.00	0.00	32.52	9,254,700.00 *gal*
Application event totals		2.99	0.00	0.00	32.52	

PVT 3 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW PVT 2 & 3	Ground water	2.85	0.00	0.00	30.97	8,814,000.00 *gal*
Application event totals		2.85	0.00	0.00	30.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW PVT 2 & 3	Ground water	2.99	0.00	0.00	32.52	9,254,700.00 *gal*
Application event totals		2.99	0.00	0.00	32.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW PVT 2 & 3	Ground water	2.99	0.00	0.00	32.52	9,254,700.00 *gal*
Application event totals		2.99	0.00	0.00	32.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW PVT 2 & 3	Ground water	3.13	0.00	0.00	34.07	9,695,400.00 *gal*
Application event totals		3.13	0.00	0.00	34.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	292.95	85.69	767.53	3,984.14	780.00 *ton*
Application event totals		292.95	85.69	767.53	3,984.14	

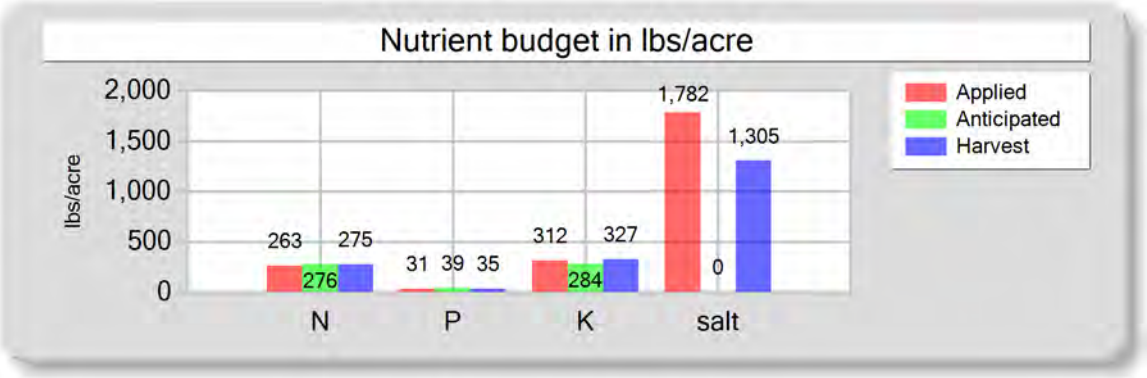
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW PVT 2 & 3	Ground water	2.85	0.00	0.00	30.97	8,814,000.00 *gal*
Application event totals		2.85	0.00	0.00	30.97	

B. NUTRIENT BUDGET

Field 4 - 01/01/2023: Pasture

Field name: Field 4 Crop: Pasture Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	221.80	30.74	312.25	1,365.86
Fresh water	26.94	0.00	0.00	416.15
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	262.74	30.74	312.25	1,782.01
Anticipated crop nutrient removal	276.00	39.00	283.50	0.00
Actual crop nutrient removal	275.40	34.50	327.00	1,305.00
Nutrient balance	-12.66	-3.76	-14.75	477.01
Applied to removed ratio	0.95	0.89	0.95	1.37

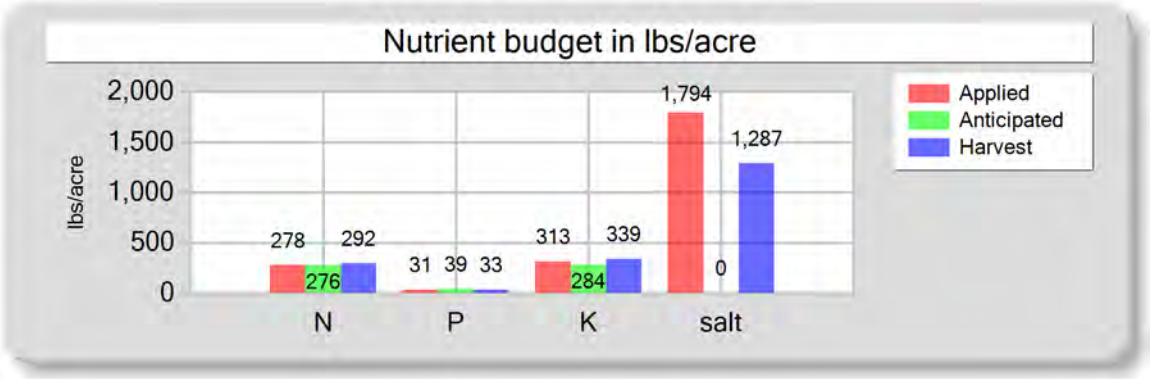
Fresh water applied	
38,988,000.00	*gallons*
1,435.80	*acre-inches*
33.39	*inches/acre*

Process wastewater applied	
2,142,000.00	*gallons*
78.88	*acre-inches*
1.83	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 01/01/2023: Pasture

Field name: Field 5 Crop: Pasture Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	234.36	30.76	313.10	1,342.04
Fresh water	29.23	0.00	0.00	451.63
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	277.59	30.76	313.10	1,793.67
Anticipated crop nutrient removal	276.00	39.00	283.50	0.00
Actual crop nutrient removal	291.90	33.00	339.00	1,287.00
Nutrient balance	-14.31	-2.24	-25.90	506.67
Applied to removed ratio	0.95	0.93	0.92	1.39

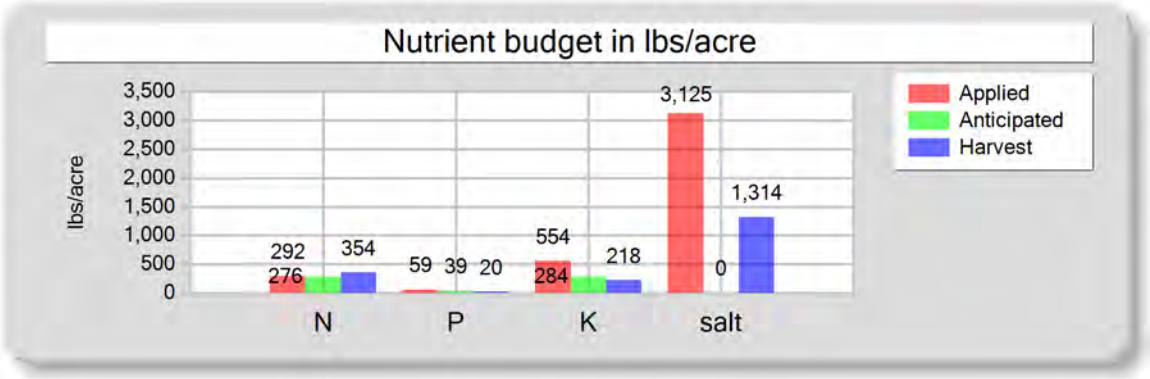
Fresh water applied	
8,856,000.00	*gallons*
326.14	*acre-inches*
36.24	*inches/acre*

Process wastewater applied	
459,000.00	*gallons*
16.90	*acre-inches*
1.88	*inches/acre*

Total harvests for the crop	
1	*harvests*

PVT 1 - 01/01/2023: Pasture

Field name: PVT 1 Crop: Pasture Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	142.72	41.75	373.93	1,940.99
Process wastewater	134.99	17.72	180.34	773.02
Fresh water	0.00	0.00	0.00	411.46
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	291.71	59.46	554.27	3,125.47
Anticipated crop nutrient removal	276.00	39.00	283.50	0.00
Actual crop nutrient removal	354.00	19.50	218.40	1,314.00
Nutrient balance	-62.29	39.96	335.87	1,811.47
Applied to removed ratio	0.82	3.05	2.54	2.38

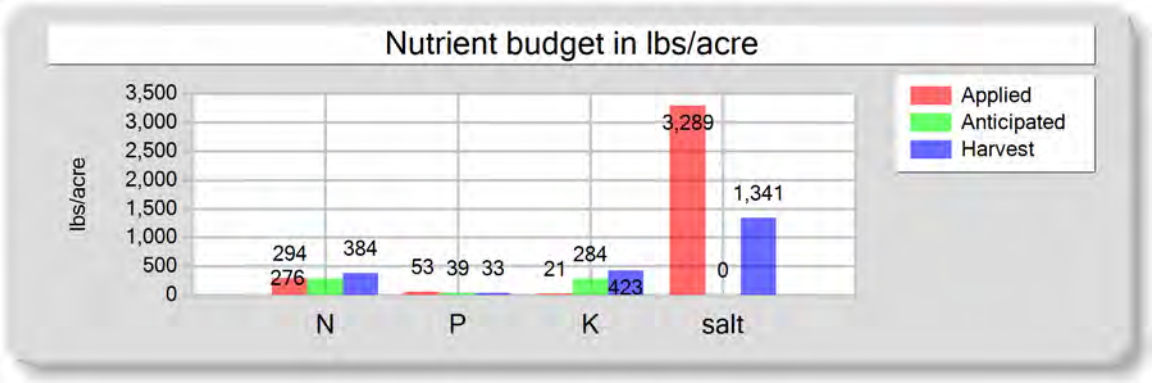
Fresh water applied	
123,265,800.00	*gallons*
4,539.46	*acre-inches*
36.32	*inches/acre*

Process wastewater applied	
3,672,000.00	*gallons*
135.23	*acre-inches*
1.08	*inches/acre*

Total harvests for the crop	
1	*harvests*

PVT 2 - 01/01/2023: Pasture

Field name: PVT 2 Crop: Pasture Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	254.94	53.20	21.06	3,014.91
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	25.22	0.00	0.00	274.08
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	294.15	53.20	21.06	3,288.99
Anticipated crop nutrient removal	276.00	39.00	283.50	0.00
Actual crop nutrient removal	384.00	33.00	423.00	1,341.00
Nutrient balance	-89.85	20.20	-401.94	1,947.99
Applied to removed ratio	0.77	1.61	0.05	2.45

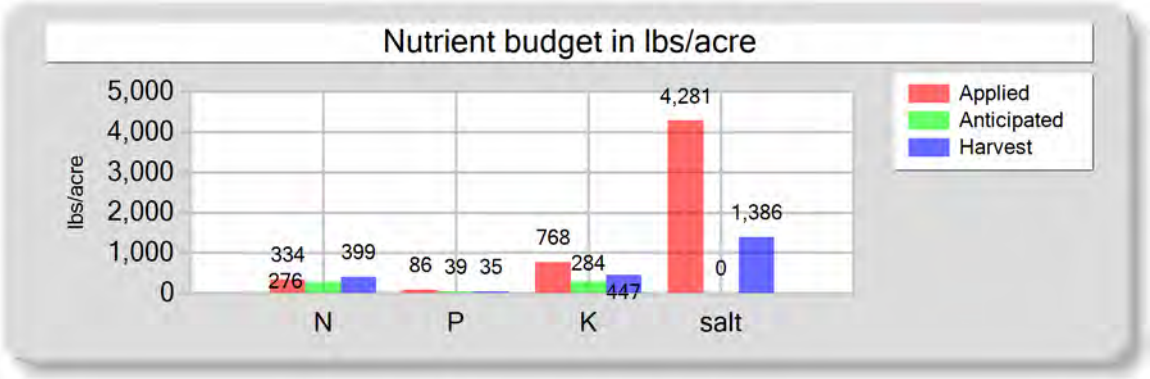
Fresh water applied	
78,003,900.00	*gallons*
2,872.62	*acre-inches*
30.24	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

PVT 3 - 01/01/2023: Pasture

Field name: PVT 3 Crop: Pasture Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	292.95	85.69	767.53	3,984.14
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	27.35	0.00	0.00	297.31
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	334.30	85.69	767.53	4,281.45
Anticipated crop nutrient removal	276.00	39.00	283.50	0.00
Actual crop nutrient removal	399.00	34.80	447.00	1,386.00
Nutrient balance	-64.70	50.89	320.53	2,895.45
Applied to removed ratio	0.84	2.46	1.72	3.09

Fresh water applied	
84,614,400.00	*gallons*
3,116.06	*acre-inches*
32.80	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids
Sample date: 04/27/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 29.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,000.00	4,800.00	1,900.00	0.00	0.00	0.00	0.00	0.00		27.20
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Solids

Sample and source description: Corral Solids
Sample date: 10/12/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 10.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,000.00	5,850.00	52,400.00	0.00	0.00	0.00	0.00	0.00		27.20
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

B. PROCESS WASTEWATER ANALYSES

WWQ1

Sample and source description: WWQ1
Sample date: 01/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	386.00	224.00	0.00	0.00	110.00	754.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5,390.00	4,120
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 05/18/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	431.00	260.00	0.00	0.00	84.00	844.00	146.00	100.00	138.00	2,660.00	0.00	147.00	119.00	5,570.00	4,080
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 07/28/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	790.00	230.00	0.00	0.00	48.80	519.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3,780.00	1,300
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 10/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	224.00	104.00	0.00	0.00	25.50	242.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2,100.00	1,320
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

C. FRESH WATER ANALYSES

IW Field 4

IW Field 4

IW Field 4

Sample description: IW Field 4

Sample date: 11/09/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.56	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	114.00	55
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

IW PVT 1

IW PVT 1

Sample description: IW PVT 1

Sample date: 11/09/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	134.00	50
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

IW PVT 2 & 3

IW PVT @ & 3

Sample description: IW PVT @ & 3

Sample date: 11/09/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.68	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	113.00	40
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field 4 - 01/01/2023: Pasture

Pasture

Sample and source description: Pasture

Sample date: 11/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,180.00	1,150.00	10,900.00		14.50
DL	100.00	100.00	100.00		0.01

Field 5 - 01/01/2023: Pasture

Pasture

Sample and source description: Pasture

Sample date: 11/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,730.00	1,100.00	11,300.00		14.30
DL	100.00	100.00	100.00		0.01

PVT 1 - 01/01/2023: Pasture

PVT 1 - 01/01/2023: Pasture

Pasture

Sample and source description: Pasture

Sample date: 11/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,200.00	1,320.00	12,600.00		15.20
DL	100.00	100.00	100.00		0.01

Pasture

Sample and source description: Pasture

Sample date: 11/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,800.00	650.00	7,280.00		14.60
DL	0.50	0.50	0.50		0.01

Pasture

Sample and source description: Pasture

Sample date: 11/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,700.00	1,020.00	10,200.00		12.50
DL	100.00	100.00	100.00		0.01

PVT 2 - 01/01/2023: Pasture

PVT 2 - 01/01/2023: Pasture

Pasture

Sample and source description: Pasture

Sample date: 11/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	950.00	12,400.00		13.80
DL	100.00	100.00	100.00		0.01

Pasture

Sample and source description: Pasture

Sample date: 11/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,800.00	1,100.00	14,100.00		14.90
DL	100.00	100.00	100.00		0.01

PVT 3 - 01/01/2023: Pasture

Pasture

Sample and source description: Pasture

Sample date: 11/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,300.00	1,160.00	14,900.00		15.40
DL	100.00	100.00	100.00		0.01

PVT 3 - 01/01/2023: Pasture

Pasture

Sample and source description: Pasture

Sample date: 11/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,300.00	880.00	14,500.00		15.20
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

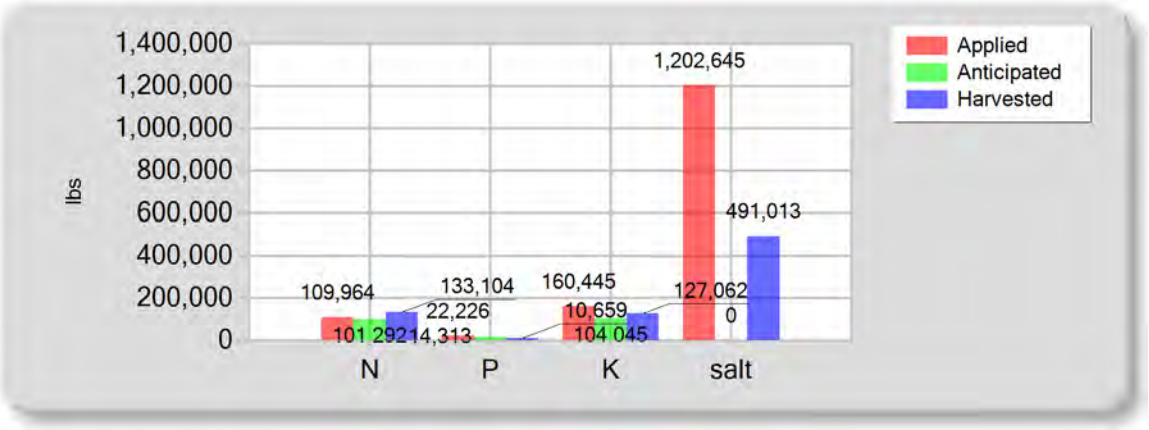
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

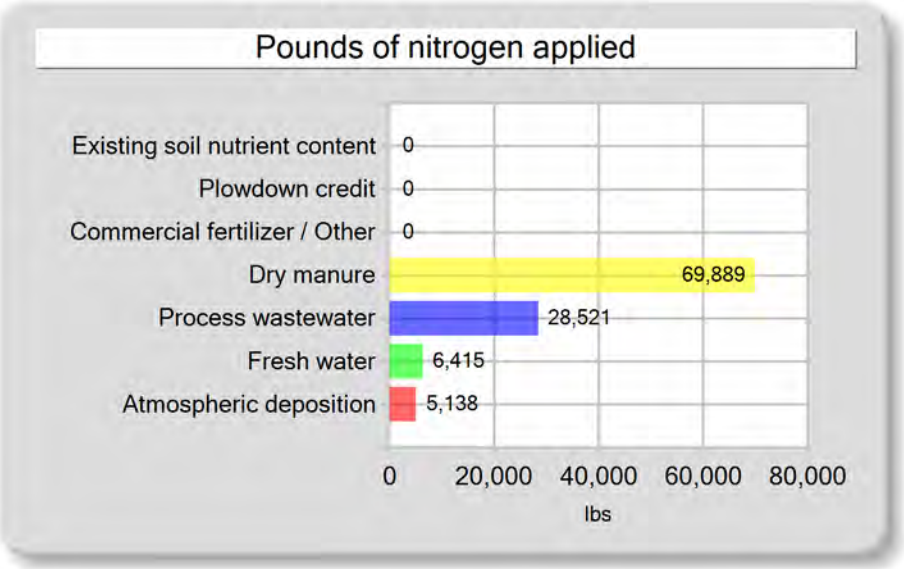
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

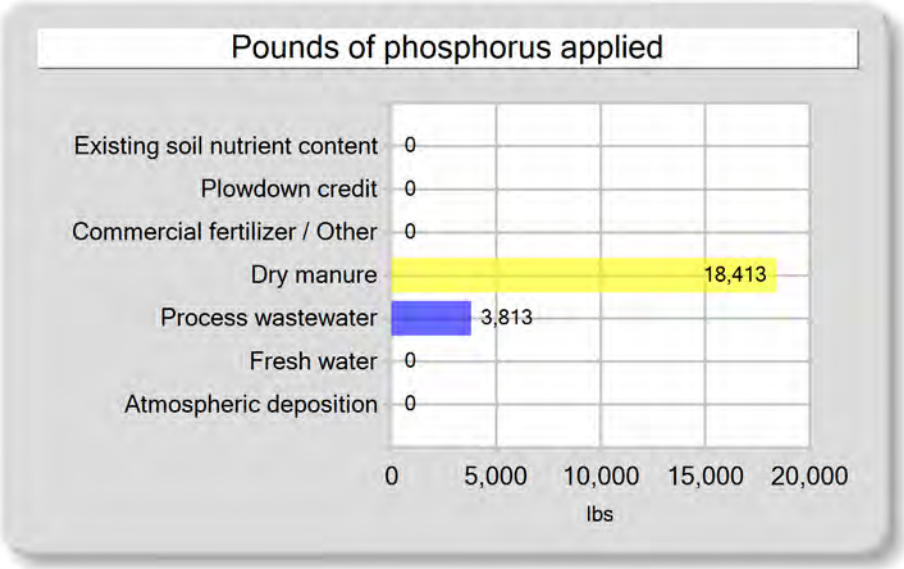
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	69,889.40	18,412.99	121,657.15	907,533.44
Process wastewater	28,520.82	3,812.99	38,787.45	167,437.58
Fresh water	6,415.30	0.00	0.00	127,673.84
Atmospheric deposition	5,138.00	0.00	0.00	0.00
Total nutrients applied	109,963.53	22,225.98	160,444.60	1,202,644.87
Anticipated crop nutrient removal	101,292.00	14,313.00	104,044.50	0.00
Actual crop nutrient removal	133,104.30	10,659.00	127,062.00	491,013.00
Nutrient balance	-23,140.77	11,566.98	33,382.60	711,631.87
Applied to removed ratio	0.83	2.09	1.26	2.45

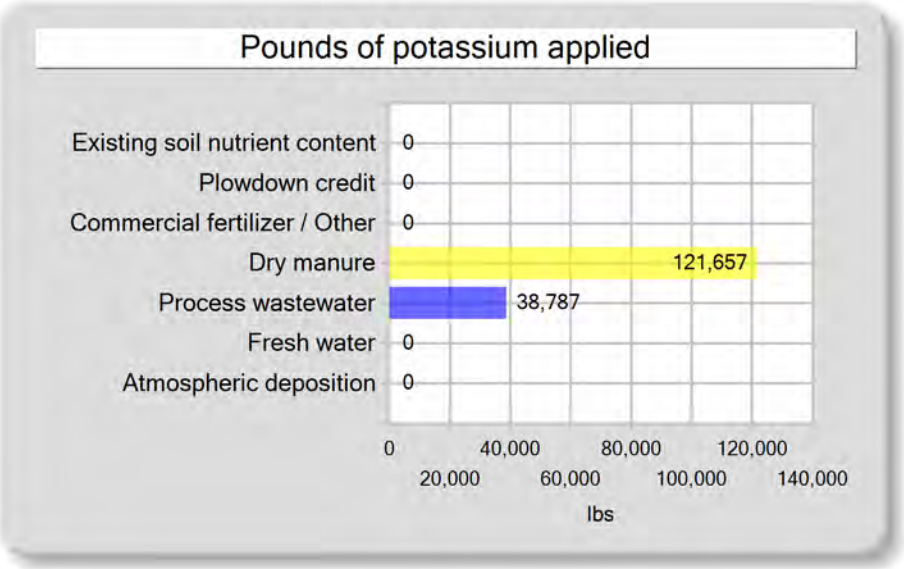
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

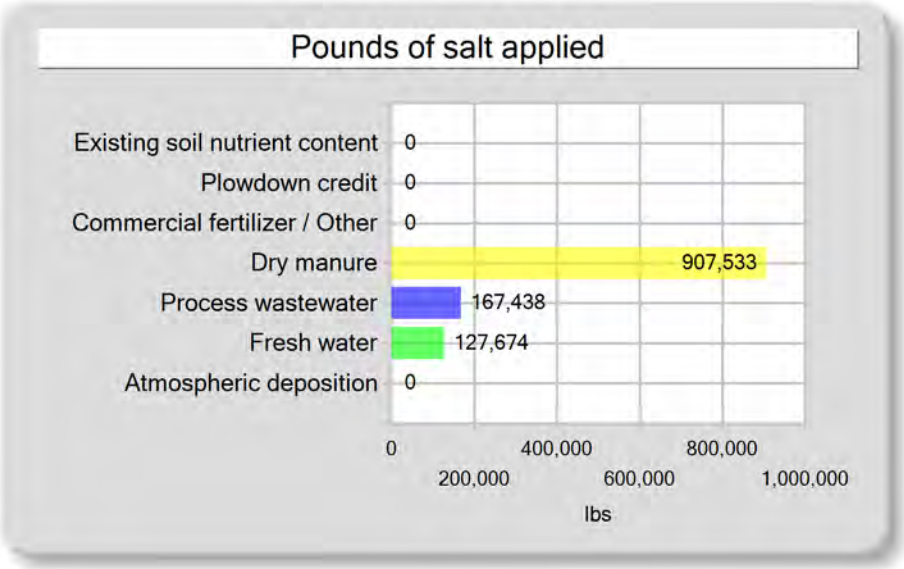


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

All milk cows were removed from the facility by 10/1/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Ward Burroughs (Apr 18, 2024 18:59 PDT)	Brian Bylsma (Apr 18, 2024 16:50 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Ward Burroughs	Brian Bylsma
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Apr 18, 2024	Apr 18, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Full Circle Dairy
6800 East Ave.
Ballico, CA 95303

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 11/9/23 2:43 PM
SUBMITTED BY: Sean
ANALYZED DATE: 11/3/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 10.4 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 1	T3313017A	Electrical Conductivity (EC)	EPA 120.1 and	0.115	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.540	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	73.4	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.6	95.8 (Method Bl)	94.0 (Method Bl)	1.89
Electrical Conductivity (EC)	95.5			1.15
Nitrate Nitrogen (NO3-N)	105	106 (T3313015A)	103 (T3313015A)	2.64

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Compliance Analysis Report

Customer
Full Circle Dairy
6800 East Ave.
Ballico, CA 95303

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 11/9/23 2:43 PM
SUBMITTED BY: Sean
ANALYZED DATE: 11/3/23

Irrigation / Ground Water (H7)
SOURCE: Well Water

NOTES: Temperature above qualifying at 10.4 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW PVT 1	T3313017B	Electrical Conductivity (EC)	EPA 120.1 and	0.134	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.41	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	50.0	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	< 1.50	mg/L
		Soluble Salts (SALT-SOL)	Calculation	85.7	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	94.6	95.8 (Method BI)	94.0 (Method BI)			1.89
Electrical Conductivity (EC)	95.5			0.491 (T3313008A)	0.486 (Dup)	1.15
Nitrate Nitrogen (NO3-N)	105	106 (T3313015A)	103 (T3313015A)			2.64
Total Dissolved Solids (TDS)	102			50.0 (T3313017B)	50.0 (Dup)	0.000
Total Nitrogen (TN)	90.7			0.000 (T3313017B)	0.000 (null)	

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Ballico, CA 95303

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 11/9/23 2:43 PM
SUBMITTED BY: Sean
ANALYZED DATE: 11/3/23

Irrigation / Ground Water (H7)
SOURCE: Well Water

NOTES: Temperature above qualifying at 11.0 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW PVT 2+3	T3313017C	Electrical Conductivity (EC)	EPA 120.1 and	0.113	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.885	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	40.0	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	3.68	mg/L
		Soluble Salts (SALT-SOL)	Calculation	72.6	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	94.6	95.8 (Method Bl)	94.0 (Method Bl)			1.89
Electrical Conductivity (EC)	95.5			0.491 (T3313008A)	0.486 (Dup)	1.15
Nitrate Nitrogen (NO3-N)	105	106 (T3313015A)	103 (T3313015A)			2.64
Total Dissolved Solids (TDS)	102			50.0 (T3313017B)	50.0 (Dup)	0.000
Total Nitrogen (TN)	90.7			0.000 (T3313017B)	0.000 (null)	

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Consultant:
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2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 11/9/23 2:43 PM
SUBMITTED BY: Sean
ANALYZED DATE: 11/3/23

Irrigation / Ground Water (H7)
SOURCE: Well Water

NOTES: Temperature above qualifying at 9.4 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW Field 4	T3313017D	Electrical Conductivity (EC)	EPA 120.1 and	0.114	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.763	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	55.0	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	3.56	mg/L
		Soluble Salts (SALT-SOL)	Calculation	72.6	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	94.6	95.8 (Method Bl)	94.0 (Method Bl)			1.89
Electrical Conductivity (EC)	95.5			0.491 (T3313008A)	0.486 (Dup)	1.15
Nitrate Nitrogen (NO3-N)	105	106 (T3313015A)	103 (T3313015A)			2.64
Total Dissolved Solids (TDS)	102			50.0 (T3313017B)	50.0 (Dup)	0.000
Total Nitrogen (TN)	90.7			0.000 (T3313017B)	0.000 (null)	

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DENELE ANALYTICAL, INC.
865 S. KILROY ROAD
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: FULL CIRCLE
Mailing address:
(not facility address) BALLICO CA
Phone#:

Authorized Copy Released To: F&R AG SERVICES
Sampled By: Sean Locke
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	11/9/23	9:00	DW 1	DOMESTIC	10.4		X				
2	|	9:20	IW PUT 1	AG WELL	10.4						H7
3	|	9:40	IW PUT 2+3	|	11.0						H7
4		10:00	IW FIELD 4	|	9.4						H7
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
		F&R AG SERVICES				
2nd		DENELE ANALYTICAL, INC.	11-9-23 @ 14:43			T3313017

Required to be submitted to any agency? (YES / NO)

Full Circle 2023 Annual Report

Final Audit Report 2024-04-19

Created:	2024-04-17
By:	Sean Locke (slocke@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAds430tTXkrr0iWjUB9lW-Yw-iMubVrkQ

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Signer admin@fullcircledairy.net entered name at signing as Brian Bylsma
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Signer ward@burroughsfamilyfarms.com entered name at signing as Ward Burroughs
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Document e-signed by Ward Burroughs (ward@burroughsfamilyfarms.com)
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